Business Requirements Document (BRD)

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Business Requirements Document – Overall (all modules) · **Version:** 1.0 · **Status:** Approved
> **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026
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Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version of the overall BRD: whole-system view, global rules, NFRs and acceptance criteria, Authentication and Legal requirements, index of the 16 sub-module BRDs. Rewrites the original BRD into the fixed template and records the changes made since.
1.0	31-Oct-2026	Jojo George	Approved and signed off; baselined as the agreed business requirements.
1.0	31-Oct-2026	Jojo George	Editorial only: Related Documents paths updated to the `TF - Documents` folder layout; no requirement changed, sign-off unchanged.

Approvals / Sign-off

Role	Name	Decision	Date
Product owner	Jojo George	Approved	31-Oct-2026
Reviewer	Jojo George	Approved	31-Oct-2026

Related Documents

Document	Location
Original BRD	`BRD for Task management.docx` (v1.0)
Development Plan (screen-wise, 20 screens)	`TF - Documents/Screens Wise Feature Development Plan/`
Sub-module BRDs (16)	See section 3.4 (links relative to this file)
FRD – Overall	`../../FRD/00-Overall/FRD-TaskFlow-Overall.md`

Document	Location
Functional / technical module workflow docs	`TaskFlow Proj/TFWorkflow/App WF in MD/` and `TaskFlow Proj/TFTechnicalWorkflow/Tech WF in MD/`
TDD / HLD / LLD	`TF - Documents/TF - TDD/TDD-TaskFlow.md`; `../../HLD/00-Overall/HLD-TaskFlow-Overall.md`; `../../LLD/00-Overall/LLD-TaskFlow-Overall.md`

How this document is organised

This overall BRD holds everything that is shared by the whole system: roles, scope, the main workflows, global business rules (`BR-nn`), global non-functional requirements (`NFR-nn`), global acceptance criteria (`AC-nn`) and the requirements of **Authentication** and **Legal**, which have no sub-module BRD. Requirements of every other screen are in the sub-module BRDs listed in section 3.4 (`REQ-<CODE>-nn`); they are summarised here in one line each and are not repeated.

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1. Project Overview

1.1 Introduction

TaskFlow is a web-based office task management application. Managers create tasks, assign them to employees, set priorities, start dates, deadlines and estimated hours, and follow progress until the work is done. Employees see their own tasks, update status, log worked hours and discuss the work in task comments. TaskFlow reminds everyone involved automatically before a deadline, on the deadline day and while a task is overdue, and gives dashboards, workload figures, activity history and reports so that nothing is missed.

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1.2 Business Problem / Current Situation

Without a shared system, tasks are handed out by word of mouth, chat or e-mail. Nobody has one list of who is doing what and by when, deadlines are remembered (or forgotten) by individuals, and managers find out about late work only when it is already late. Workload is invisible, so some people are overloaded while others wait. Changes to a task (new deadline, new assignee) are not recorded, and reports have to be counted by hand. Access to information is not controlled: anyone with a password can see everything, and a stolen password is enough to get in.

1.3 Project Objectives

ID	Objective	Success measure
OBJ-01	Simplify employee task allocation.	A manager creates and assigns a task on one screen; the assignee is notified as soon as it is saved.
OBJ-02	Monitor the progress of assigned tasks.	Every task has exactly one of six statuses and the same status is shown in the Task List, the Board and the dashboards.
OBJ-03	Reduce missed deadlines through automatic reminders.	With the default settings every open task gets reminders 2 days and 1 day before the deadline, on the deadline day and every day while overdue, without anyone acting.
OBJ-04	Notify the assignee and selected recipients about approaching deadlines.	The assignee, each additional recipient and the task creator receive each reminder exactly once.
OBJ-05	Help managers identify pending and overdue work and workload imbalance.	Dashboards and Employee Workload show pending, overdue and high-load figures for the user's scope; a click opens the matching tasks.
OBJ-06	Improve communication between managers and employees.	Task comments, in-app notifications and internal Mail are available inside TaskFlow; nothing has to be sent outside it.
OBJ-07	Maintain a history of task updates and changes.	Every important task change is recorded with who made it, when, and the old and new value.
OBJ-08	Generate reports for better work planning.	Authorised users filter task totals and export the filtered tasks to CSV.
OBJ-09	Protect access to the organisation's data.	No one uses TaskFlow without a password and an authenticator code; each user sees only the data their permissions allow; old sessions end after any security change.
OBJ-10	Let the organisation configure TaskFlow without changing code.	Reminder defaults, timezone, default priority, notifications, roles and the AI provider are changed from Settings screens.
OBJ-11	Give users self-service help with their work.	TFA answers "how do I...?" questions and prepares actions that run only after the user confirms them.
OBJ-12	Make ownership, terms of use and privacy clear to everyone.	The five legal documents can be opened from the footer of every main screen, also before sign-in.

1.4 Expected Benefits

- **Organisation:** fewer missed deadlines, one reliable record of all work and its history, and reports without manual counting.
- **Administrators:** one place to manage people, departments, projects, roles and settings; access is controlled by permissions and protected by two-step verification.

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- **Managers:** quick task assignment, early warning of overdue and delayed work, visible team workload and a recorded reason for every reassignment.
- **Employees:** a personal list of their own work and deadlines, automatic reminders, and a place to discuss each task.
- **All users:** self-service password recovery and profile updates, internal Mail, and an assistant that explains how TaskFlow works.

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2. Stakeholders & User Roles

2.1 Stakeholders

Stakeholder	Interest / responsibility
Project owner (Jojo George)	Defines priorities, owns TaskFlow and its legal documents, approves this BRD.
Administrators	Maintain people, departments, projects, roles and system settings; help users who lost their authenticator.
Managers	Assign and follow up work, reassign tasks, review workload and reports, set reminder defaults.
Employees	Carry out assigned work, update status and hours, discuss tasks, act on reminders.
Custom-role users	People whose access is tailored by an administrator from the permission catalogue.
Visitors (not signed in)	Sign in, recover a password, read the legal documents.

2.2 User Roles

Role	Description	Main responsibilities	Access level
Administrator	System role holding every permission in the catalogue.	Manage employees (including MFA reset), departments, projects and positions; all settings and roles; all tasks and reports.	Organisation (full system)
Manager	System role for people who lead work.	Create, edit, reassign tasks and manage recipients; view workload, reports and project details; manage reminder settings.	Team for tasks and the team dashboard; organisation for reports and workload (see the sub-module BRDs)
Employee	System role for people who carry out work.	View own tasks, change status, log hours, comment; use notifications, Mail, profile and TFA.	Own records only
Custom Role	Role created by an administrator from the fixed permission catalogue.	Whatever the granted permissions allow.	As granted: Own, Team or Organisation
Visitor	Anyone who is not signed in.	Sign in, recover a password, read legal documents.	Public pages only

The system roles (Administrator, Manager, Employee) cannot be changed. Every user holds exactly one role.

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2.3 Default Permissions by Module (summary)

The permission catalogue has 64 permissions (see the sub-module BRDs for each key). "Always On" permissions belong to every role and cannot be removed; custom roles receive only what an administrator grants.

Module	Administrator	Manager	Employee	Custom Role
Authentication, Legal	Public (no permission)	Public	Public	Public
Dashboard (5)	All	Own and team dashboard, analytics, filters	Own dashboard, analytics, filters	Own dashboard Always On; others configurable
TaskFlow Monitoring (17)	All	All except delete task	Own tasks, status change, comments, activity history, board	Own tasks Always On; others configurable
TaskFlow Management (18)	All	View employees, departments and projects only	None	Configurable
Communication (6)	All	All (own mailbox and notifications)	All (own mailbox and notifications)	Configurable
Reports (2)	View and export	View and export	None	Configurable
Settings (10)	All	Reminder settings only	None	Configurable; "give the Administrator role" is never assignable
My Profile (4)	Always On	Always On	Always On	Always On
TFA Assistant (2)	Chat and confirm actions	Chat and confirm actions	Chat and confirm actions	Configurable

3. Scope

3.1 In Scope

- Sign-in with email or username and password, mandatory two-step verification with an authenticator app and backup codes, self-service password recovery, secure sessions and sign-out.
- Role-specific dashboards with analytics, filters and drill-down to the matching tasks.
- Employee, department and project management, including positions and project allocation percentages.
- Task creation and assignment, status tracking with fixed transitions, deadlines and estimated hours, worked hours, notification recipients, reassignment with a reason, soft delete.
- Task comments and a full activity history.
- Kanban board and employee workload.
- In-app notifications, automatic deadline reminders, and internal Mail between TaskFlow users.
- Task reports with filters and CSV export.

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- Application settings, reminder settings, role configuration with custom roles, and TFA (AI provider) settings.
- My Profile, change password and self-service two-step verification.
- TFA, the TaskFlow Assistant, with confirm-before-acting actions, help topics, audit log, rate limits and voice input.
- Legal documents and the ownership footer.

3.2 Out of Scope (this release)

- Sending e-mail or SMS outside TaskFlow (SMTP configuration of the original Screen 14) – superseded by internal Mail; nothing leaves TaskFlow.
- Several organisations (tenants) in one installation – TaskFlow serves one organisation.
- Running more than one application server at the same time – a single instance is assumed.
- Automatic scheduled backups – backups are an operational task with a documented procedure.
- Editing legal documents from the application – the content is fixed in the application.
- Password reset by e-mail link or by an administrator choosing a password – recovery uses the authenticator app; an administrator can only reset two-step verification.

3.3 Modules Overview

#	Module	Purpose	Screens	Specified in
1	Authentication	Sign in, two-step verification, account recovery, security setup, sessions	Login, Account Recovery, Security Setup	This document, SCREEN 1–3
2	Dashboard	Role-specific summary and analytics	Administrator, Manager, Employee Dashboard	BRD-Dashboard
3	TaskFlow Monitoring	Tasks, comments, history, board, workload	Task List, Create Task, Task Details, Reassign Task, Task Discussion, Activity History, Board, Employee Workload	BRD-Task-Management, BRD-Board, BRD-Employee-Workload
4	TaskFlow Management	Master data: people, departments, projects	Employee Management, Department Management, Project Management, Project Details	BRD-Employee-Management, BRD-Department-Management, BRD-Project-Management
5	Communication	Notifications, reminder engine, internal Mail	Notification Center, Mail	BRD-Notification, BRD-Mail
6	Reports	Task totals and CSV export	Task Reports	BRD-Reports
7	Settings	Organisation-wide configuration	Reminder, Application, TFA Settings, Role Configuration	BRD-Reminder-Settings, BRD-Application-Settings, BRD-TFA-Settings, BRD-Role-Configuration
8	My Profile	Own account, password, two-step verification	My Profile, Change Password	BRD-My-Profile
9	TFA Assistant	AI assistant panel on every signed-in screen	TFA Panel, Action Card	BRD-Assistant

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#	Module	Purpose	Screens	Specified in
10	Legal	Legal documents and ownership footer	Five legal pages, legal footer	This document, SCREEN 4–5

3.4 Module Index – Sub-module BRDs

Each sub-module BRD uses its code in its IDs (`OBJ-`, `REQ-`, `BR-`, `NFR-`, `R-`, `AC-<CODE>-nn`) and refers to the global rules of section 6 of this document in words.

#	Code	Module	Sub-module	Sub-module BRD
0	ALL	(all)	Overall, incl. Authentication and Legal	This document
1	DASH	Dashboard	Dashboard	BRD-Dashboard
2	TASK	TaskFlow Monitoring	Task Management (incl. comments and activity history)	BRD-Task-Management
3	WORK	TaskFlow Monitoring	Employee Workload	BRD-Employee-Workload
4	BOARD	TaskFlow Monitoring	Board	BRD-Board
5	EMP	TaskFlow Management	Employee Management	BRD-Employee-Management
6	PROJ	TaskFlow Management	Project Management (incl. positions)	BRD-Project-Management
7	DEPT	TaskFlow Management	Department Management	BRD-Department-Management
8	MAIL	Communication	Mail	BRD-Mail
9	NOTIF	Communication	Notification (incl. reminder engine)	BRD-Notification
10	RPT	Reports	Reports	BRD-Reports
11	REM	Settings	Reminder Settings	BRD-Reminder-Settings
12	APP	Settings	Application Settings	BRD-Application-Settings
13	TFAS	Settings	TFA Settings	BRD-TFA-Settings
14	ROLE	Settings	Role Configuration	BRD-Role-Configuration
15	PROF	My Profile	My Profile (profile, change password, two-step verification)	BRD-My-Profile
16	TFA	TFA Assistant	Assistant	BRD-Assistant

Authentication requirements in this document use `REQ-AUTH-nn`; Legal requirements use `REQ-LEGAL-nn`.

3.5 Change from Original BRD (system level)

Original BRD item	Current requirement	Detailed in
Screen 1 – password-only login	Password plus mandatory two-step verification (authenticator app, 10 backup codes); self-service recovery with an authenticator code	SCREEN 1–3

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Original BRD item	Current requirement	Detailed in
Fixed roles Administrator / Manager / Employee	Permission-based access from a fixed catalogue; read-only system roles plus custom roles	BRD-Role-Configuration
Screen 8 "initial status"	Every task starts as To Do; status changes follow fixed transitions	BRD-Task-Management
Task statuses To Do, In Progress, Blocked, Completed, Cancelled	**In QA** added; the Board has five columns (To Do, In Progress, In QA, Blocked, Completed)	BRD-Task-Management, BRD-Board
Screen 14 – Email Notification Settings (SMTP)	**Superseded** by internal Mail; reminders can be copied into the recipient's Mail inbox ("Also deliver reminders to Mail" on Reminder Settings)	BRD-Mail, BRD-Reminder-Settings
Reminder recipients = task recipients	Task recipients (assignee always one) **plus the task creator**, active users only	4.2, BRD-Notification
Fixed reminder schedule	First and second reminder per task (0–30 days), due-today, overdue Daily or Weekly; defaults in Reminder Settings	BRD-Reminder-Settings
Screen 19 "require the current password"	Password change confirmed with an authenticator or backup code; signs the user out everywhere	BRD-My-Profile
Screen 20 default reminder intervals	Kept on Reminder Settings, linked from Application Settings	BRD-Application-Settings
Department filters on Screens 5, 16, 17	Project filters; projects with positions and allocation percentages added	BRD-Employee-Workload, BRD-Reports
Not in the original BRD	Projects and positions, departments with managers, Role Configuration, TFA assistant and TFA Settings, internal Mail, Security Setup, Account Recovery, Legal pages	Section 3.4

4. Business Process / Workflows

4.1 Task Lifecycle Workflow

Trigger: A user with "create tasks" saves a new task · **Actors:** Manager or Administrator, Employee (assignee), recipients, System

Step	Actor / System	Action	Result
1	Manager	Creates a task with title, project, assignee, priority, start date, deadline, estimated hours, optional recipients and reminder days	Task saved as **To Do** with a unique code (e.g. `TSK-000042`); creator and creation time recorded
2	System	Records "Task created" and "Assigned to ..." in the activity history and notifies the assignee	Assignee sees the task in My Tasks, the Board and the dashboard
3	Employee	Moves the task to **In Progress** (Task Details or Board) and logs worked hours	Status change and hours recorded; recipients notified of hours
4	Employee / Manager	Discuss the task in comments	Comments recorded and notified; history updated

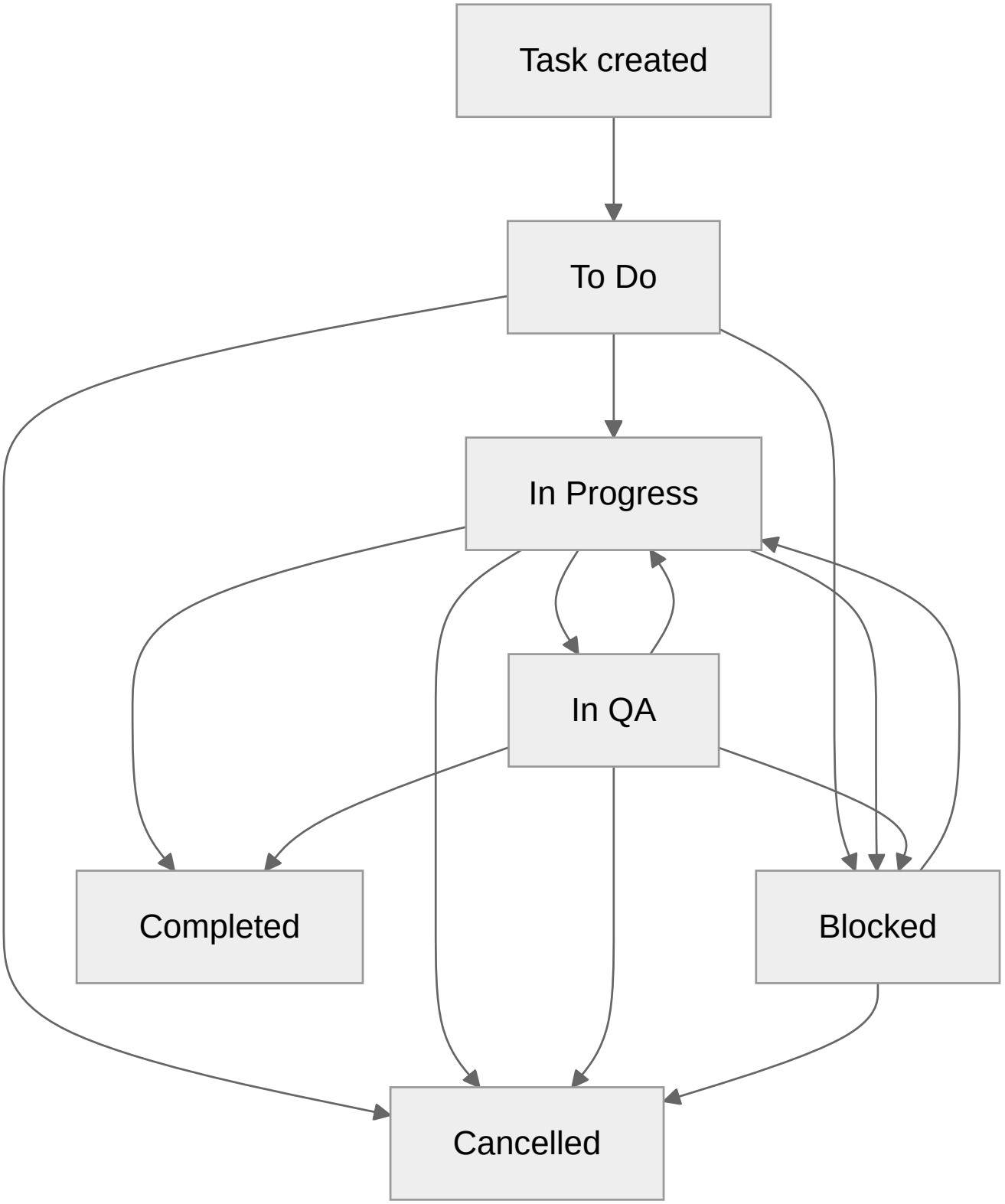
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Step	Actor / System	Action	Result
5	Employee	Moves the task to **In QA** when the work is ready to be checked, or to **Blocked** when it cannot continue	Status recorded in the history
6	Manager	Reassigns the task if needed, with a reason	Previous and new assignee recorded; new assignee notified; history kept
7	Employee / Manager	Moves the task to **Completed** (or **Cancelled**)	Completion (or cancel) time stored; reminders stop; the status is final

Allowed status moves: To Do → In Progress, Blocked or Cancelled; In Progress → In QA, Blocked, Completed or Cancelled; In QA → In Progress, Blocked, Completed or Cancelled; Blocked → In

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Progress or Cancelled. Completed and Cancelled are final. Overdue is not a status: a task is overdue while its deadline has passed and it is neither Completed nor Cancelled.



Example: On 10-Oct-2026 the manager Anita creates "Prepare Monthly Report" (Project: Finance Ops, priority High, start 10-Oct-2026 09:00, deadline 15-Oct-2026 17:00, estimate 8 hours) for Rahul, with Priya as additional recipient. TaskFlow saves it as `TSK-000042`, To Do. Rahul moves it to In Progress on 12-Oct and logs 3 hours; on 14-Oct he moves it to In QA; Anita checks it and moves it to

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Completed on 15-Oct at 11:00. The history shows every step with user and time, and no further reminders are sent.

Exceptions / alternate flows:

- A move that is not allowed (for example Completed → In Progress) is refused and the reason is shown.
- An inactive employee cannot be chosen as assignee; a deadline before the start date or an estimate of 0 hours is refused.
- Deleting a task (Administrator by default) hides it from every list but keeps its comments and history.

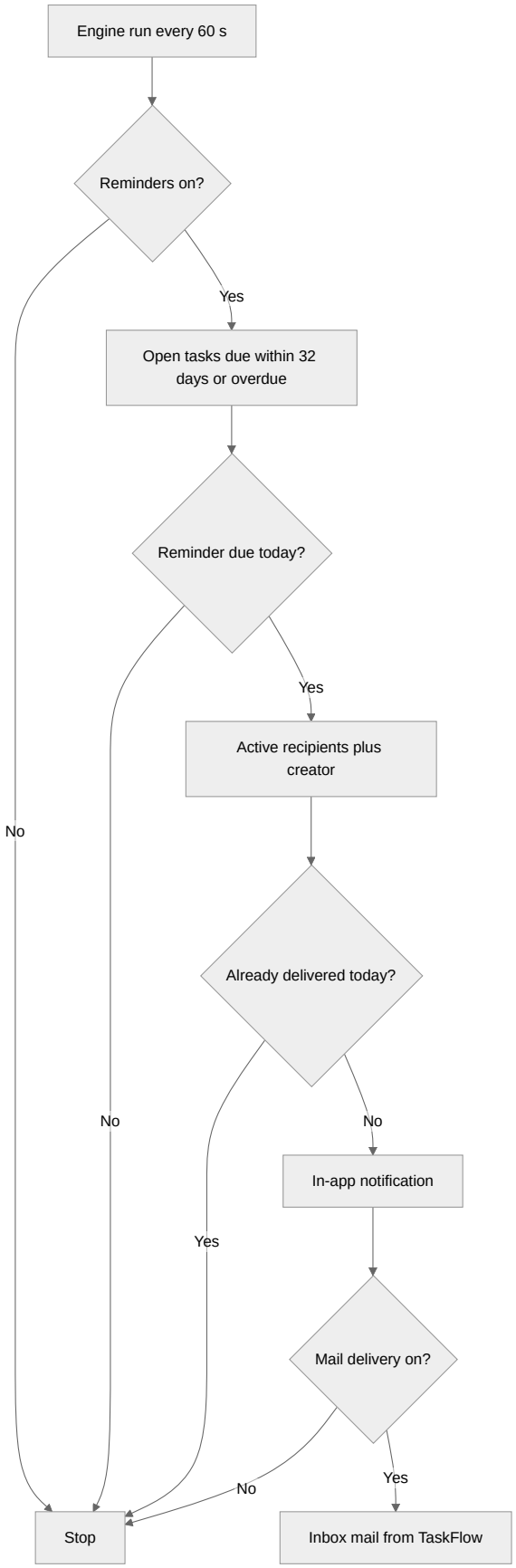
Details: BRD-Task-Management, BRD-Board.

4.2 Automatic Deadline Reminder Workflow

Trigger: The reminder engine runs when the application starts and then every 60 seconds · **Actors:** System, assignee, recipients, task creator

Step	Actor / System	Action	Result
1	System	Checks whether "Send deadline reminders" is on in Reminder Settings	Off: nothing happens
2	System	Finds every open task (not Completed, Cancelled or deleted) due within 32 days or already overdue	List of candidate tasks
3	System	Decides, using the application timezone, whether a reminder is due **today**: first reminder, second reminder, due today (before the deadline time), or overdue (daily or weekly)	Reminder type for today, or none
4	System	Works out the recipients: the task's notification recipients (the assignee is always one) plus the creator, active users only	Recipient list
5	System	Skips any reminder already delivered today for that task, person and type	No duplicates
6	System	Creates the in-app notification; when "Also deliver reminders to Mail" is on, also puts a mail from "TaskFlow" in the person's Inbox	Recipients are reminded
7	System	If the task was completed, cancelled, deleted or given a new deadline in the meantime, writes nothing	Reminders follow the current task

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Example: Default settings (first reminder 2 days, second 1 day, deadline-day reminder on, overdue Daily, Mail on). Task "Prepare Monthly Report" for Rahul, created by Anita, extra recipient Priya, deadline 15-Oct-2026 17:00 (Asia/Kolkata). On 13-Oct Rahul, Priya and Anita each receive the first reminder; on 14-Oct the second; on 15-Oct, before 17:00, the due-today reminder; from 16-Oct an

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overdue reminder every day while the task stays open. Each reminder appears under the bell and in each person's Mail inbox exactly once, even though the engine runs every minute and the server is restarted on 14-Oct. When Rahul completes the task on 16-Oct, the overdue reminders stop.

Exceptions / alternate flows:

- A day that has already ended is never back-filled; only reminders due today are sent.
- If the first and second reminder fall on the same day, only one reminder is sent that day.
- Inactive or deleted users receive no reminders.
- A failed run is logged and the next run continues.

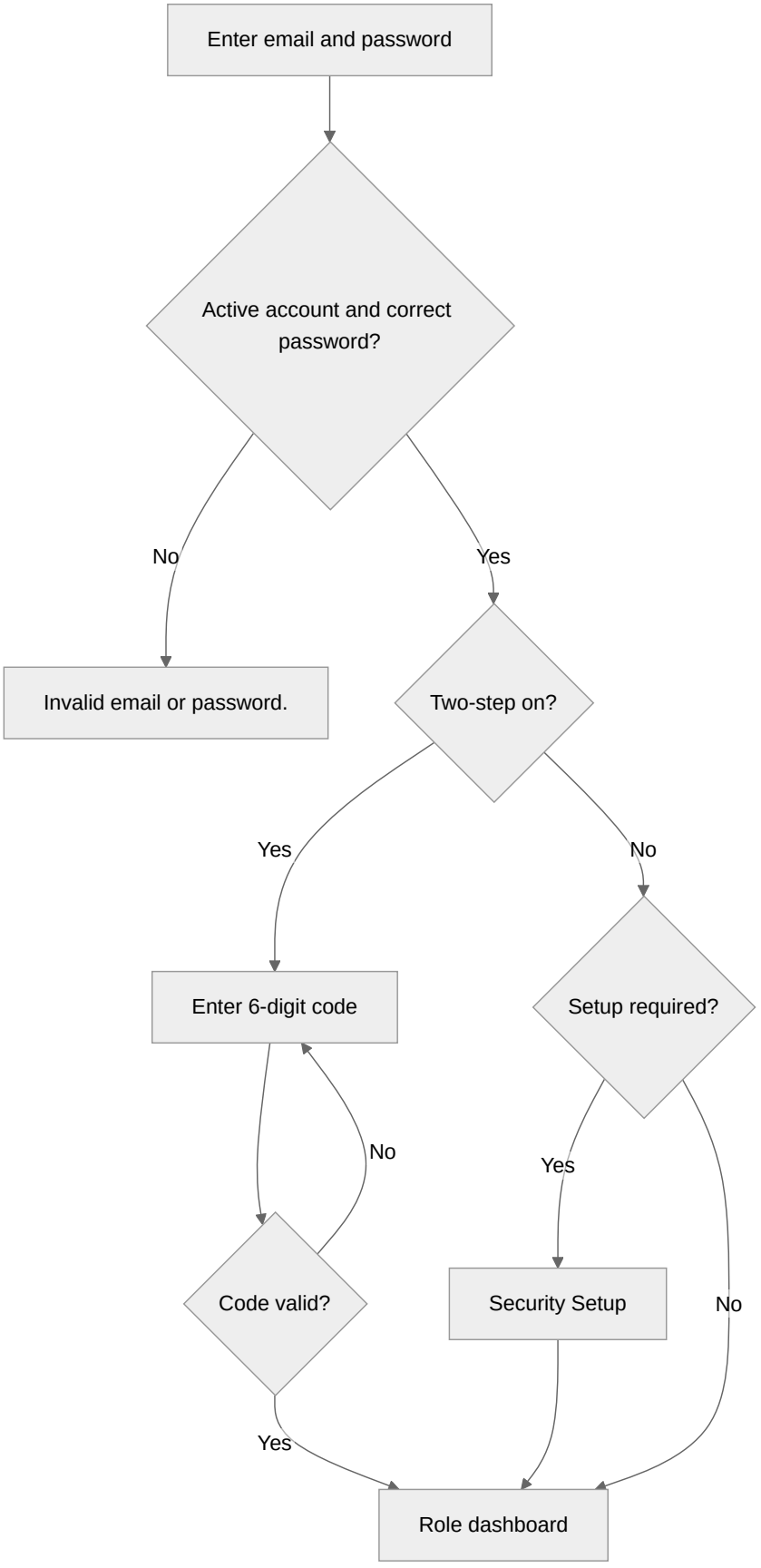
Details: BRD-Notification, BRD-Reminder-Settings.

4.3 Sign-in with Two-Step Verification (MFA) Workflow

Trigger: A user opens TaskFlow and signs in · **Actors:** Any registered user, System

Step	Actor / System	Action	Result
1	User	Enters email (or username) and password on the Login screen	–
2	System	Checks that the account exists, is active and the password is correct	Any failure: "Invalid email or password."
3	System	If two-step verification is on, asks for the 6-digit code instead of opening a session	Code step shown; the sign-in step is valid for 5 minutes
4	User	Enters the current code from the authenticator app	Correct, unused code: session opens. 5 wrong codes: locked for 5 minutes
5	System	If two-step verification is required but not yet set up, opens Security Setup and blocks every other screen	User must finish setup
6	User	Security Setup: confirms details, re-enters password, scans the QR code, enters the first code, saves 10 backup codes	Two-step verification is on
7	System	Opens the user's dashboard (Administrator, Manager, or personal; custom roles by their dashboard permissions)	User works in TaskFlow

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Example: Priya's account is created by the administrator. At her first sign-in on 20-Oct-2026 she enters `priya@taskflow.com` and her password; TaskFlow opens "Let's secure your account". She confirms her mobile number +919876543210, re-enters her password, scans the QR code with

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Microsoft Authenticator, enters 482913 and downloads her 10 backup codes. She lands on her personal dashboard. The next day she signs in with her password and the current code.

Exceptions / alternate flows:

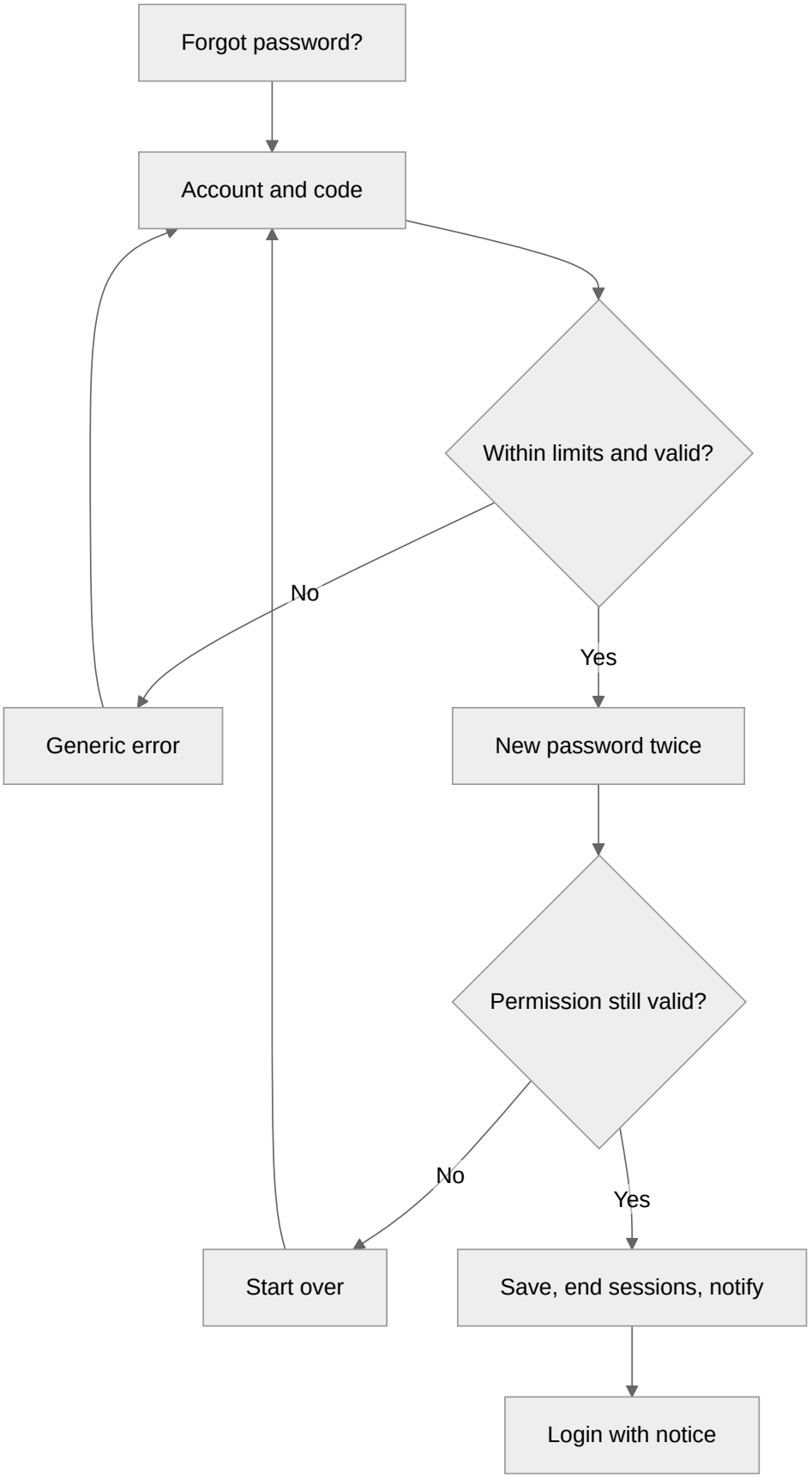
- A code that was already used, or a sign-in step older than 5 minutes, is refused: "The code is incorrect, or the sign-in has expired. Try again or sign in again."
- A user who forgot the password uses Account Recovery (SCREEN 2) with an authenticator code.
- A user who lost both phone and backup codes asks an administrator to reset two-step verification (Employee Management); all of that user's sessions end.

4.4 Self-Service Account Recovery Workflow

Trigger: A user clicks **Forgot password?** on the Login screen · **Actors:** User with two-step verification on, System

Step	Actor / System	Action	Result
1	User	Enters email or username and the current authenticator code	–
2	System	Checks attempt limits, account, two-step status and code	Any failure: one generic message
3	System	Grants a single-use reset permission valid for 10 minutes	New-password step shown
4	User	Enters and confirms a new password (6–128 characters)	–
5	System	Saves the password, ends every session of the user, cancels other reset permissions, sends a System notification	User signs in with the new password

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Example: On 22-Oct-2026 Rahul forgets his password. He enters `rahul` and the code 905117, chooses a new password and confirms it within 10 minutes. His laptop session that was still open is signed out, he receives the notification "Your password was reset with your authenticator app. If this was not you, contact an administrator now." and signs in again with the new password.

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Exceptions / alternate flows:

- More than 5 attempts for one account, or 20 from one network address, within 5 minutes: "Too many attempts. Try again in a few minutes."
- Users without two-step verification must ask an administrator.

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5. Screen-by-Screen Requirements

5.1 Screen Map (user-journey order)

Full requirement blocks are given below only for Authentication and Legal screens (no sub-module BRD). Every other screen is specified in the sub-module BRD named in the last column.

#	Screen	Module	Access	Original BRD screen	Specified in
1	Login (with two-step verification step)	Authentication	Everyone (public)	1	SCREEN 1 below
2	Account Recovery	Authentication	Everyone (public); works with two-step on	– (new)	SCREEN 2 below
3	Security Setup	Authentication	Signed-in users with setup pending	– (new)	SCREEN 3 below
4	Administrator Dashboard – organisation summary and analytics	Dashboard	Administrator; `dashboard.view.organization`	2	BRD-Dashboard
5	Manager Dashboard – team summary, delayed work, team workload	Dashboard	Manager; `dashboard.view.team`	3	BRD-Dashboard
6	Employee Dashboard – own tasks, deadlines, projects	Dashboard	Every signed-in user	4	BRD-Dashboard
7	Task List – search, filter and page through accessible tasks	TaskFlow Monitoring	All signed-in users (scope by permission)	7	BRD-Task-Management
8	Create Task – create, assign or copy a task	TaskFlow Monitoring	`tasks.create`	8	BRD-Task-Management
9	Task Details – full information, edit, status, hours	TaskFlow Monitoring	Users who can access the task	9	BRD-Task-Management
10	Reassign Task – new assignee with a reason	TaskFlow Monitoring	`tasks.reassign`	15	BRD-Task-Management
11	Task Discussion – post, edit, delete comments	TaskFlow Monitoring	Users who can access the task	11	BRD-Task-Management
12	Activity History – filtered change history	TaskFlow Monitoring	`tasks.activity.view`	18	BRD-Task-Management

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#	Screen	Module	Access	Original BRD screen	Specified in
13	Board – status columns with drag and drop	TaskFlow Monitoring	`board.view`	10	BRD-Board
14	Employee Workload – task counts and allocation per employee	TaskFlow Monitoring	`workload.view`	16	BRD-Employee-Workload
15	Employee Management – list, details panel (incl. Reset MFA), add/edit	TaskFlow Management	Administrator; employee action permissions	5	BRD-Employee-Management
16	Department Management – list, add/edit, manage members	TaskFlow Management	Administrator; department action permissions	6	BRD-Department-Management
17	Project Management and Project Details – projects, teams, positions	TaskFlow Management	Administrator; Manager read only; `projects.view`	– (new)	BRD-Project-Management
18	Notification Center (with header bell) – own notifications	Communication	`notifications.view`	12	BRD-Notification
19	Mail and Compose panel – private internal mailbox	Communication	`mail.read`, `mail.compose`	14 (superseded, replaced by Mail)	BRD-Mail
20	Task Reports – filtered totals and CSV export	Reports	`reports.view`, `reports.export`	17	BRD-Reports
21	Reminder Settings – reminder on/off, defaults, Mail delivery	Settings	`settings.reminders.manage`	13	BRD-Reminder-Settings
22	Application Settings – name, timezone, default priority, notifications	Settings	`settings.application.manage`	20	BRD-Application-Settings
23	TFA Settings – AI model, server, keys, connection test	Settings	`settings.ai.view`	– (new)	BRD-TFA-Settings
24	Role Configuration – roles and permissions	Settings	`roles.manage`	– (new)	BRD-Role-Configuration
25	My Profile, Change Password, Two-Step Verification (self-service)	My Profile	All signed-in users (Always On)	19	BRD-My-Profile
26	TFA Panel, How-to Help, Action Card, Voice Controls	TFA Assistant	`assistant.chat`, `assistant.actions.confirm`	– (new)	BRD-Assistant
27	Legal Document Pages (Copyright, Terms, Privacy, Disclaimer, Licences)	Legal	Everyone (public)	– (new)	SCREEN 4 below
28	Legal Footer and Ownership Notice	Legal	Everyone	– (new)	SCREEN 5 below

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SCREEN 1: Login (with Two-Step Verification step)

Field	Value
Screen name	Login (Sign in) and Two-step verification
Module	Authentication
Access	Everyone who is not signed in (Visitor); completed by Administrator, Manager, Employee and Custom Role users with an active account
Priority	Must

Purpose Let a registered, active user sign in securely, with the authenticator code as a second step, and open the dashboard that fits the user's role. The session opened here is kept alive and ended safely.

Screen Elements

- TaskFlow logo ("TF"), title **TaskFlow** and subtitle "Sign in to continue".
- Notice area explaining why the user was sent back to sign in.
- **Email** field (pre-filled with the identifier last used on this device) and **Password** field with **Show password / Hide password**.
- Error message area, **Sign in** button ("Signing in..." while working), **Forgot password?** link.
- Two-step step: title **Two-step verification**, "Enter the 6-digit code from your authenticator app.", **Authentication code** field, **Verify** button, **← Back to sign in** link.
- **Sign out** icon in the header account area of every signed-in screen.
- Legal footer.

Functional Requirements

ID	Requirement	Priority
REQ-AUTH-01	The system shall let a user sign in with their username or email address and their password.	Must
REQ-AUTH-02	The system shall show the same message "Invalid email or password." for an unknown account, an inactive or deleted account and a wrong password.	Must
REQ-AUTH-03	The system shall store passwords only as secure hashes and upgrade an old hash on the next successful sign-in.	Must
REQ-AUTH-04	The system shall not open a session after a correct password when two-step verification is on, but ask for the 6-digit authenticator code.	Must
REQ-AUTH-05	The system shall keep the sign-in step valid for 5 minutes and invalidate it earlier when the account's password, recovery or MFA reset changes.	Must
REQ-AUTH-06	The system shall accept each authenticator code only once, together with codes from one 30-second step before or after the current one.	Must

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ID	Requirement	Priority
REQ-AUTH-07	The system shall lock code checking for 5 minutes after 5 wrong codes in a row.	Must
REQ-AUTH-08	The system shall open, after sign-in, the page originally asked for if it is inside TaskFlow, otherwise the dashboard for the user: Administrator → Administrator Dashboard, Manager → Manager Dashboard, custom roles → the widest dashboard their permissions allow, everyone else → the personal dashboard.	Must
REQ-AUTH-09	The system shall send a user whose two-step verification is required but not set up to Security Setup after sign-in.	Must
REQ-AUTH-10	The system shall record the time of the last successful sign-in.	Should
REQ-AUTH-11	The system shall let the user show and hide the typed password.	Should
REQ-AUTH-12	The system shall pre-fill the Email field with the identifier last used on the same device.	Could
REQ-AUTH-13	The system shall keep the user signed in across page reloads with a 30-minute access pass that is renewed silently using a 7-day renewal cookie that page scripts cannot read.	Must
REQ-AUTH-14	The system shall replace the renewal cookie at every renewal so that a used, expired or revoked cookie cannot be used again.	Must
REQ-AUTH-15	The system shall check the session every 15 seconds and when the browser tab becomes visible, and apply role and permission changes without a new sign-in.	Must
REQ-AUTH-16	The system shall refuse, on every request, a session whose user is inactive or deleted or whose security stamp has changed.	Must
REQ-AUTH-17	The system shall end the session and revoke the renewal cookie when the user clicks **Sign out**, and open the Login screen.	Must
REQ-AUTH-18	The system shall show the Login screen with a notice explaining the reason when a session ends.	Should

Validations & Messages

Situation	Rule / message
Email empty	"Email is required."
Email not valid	"Enter a valid email address."
Password empty	"Password is required."
Identifier or password too long	Server accepts at most 256 characters for the identifier and 128 for the password.
Wrong credentials / inactive or deleted account	"Invalid email or password."
Code empty	"Enter the code to continue."
Wrong or expired code	"The code is incorrect, or the sign-in has expired. Try again or sign in again."

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Situation	Rule / message
Too many wrong codes	"Too many wrong codes. Try again in a few minutes."
Code failed (no server message)	"That code did not work. Check the code and try again."
Session ended or renewal failed	"Your session has ended, for example because your username or password was changed. Please sign in again."
Credentials changed	"Your username or password was changed. Sign in again with your new details."
After password reset	"Your password has been reset. Sign in with your new password."

> ⚠️ To be confirmed: the Login form asks for a valid email address although the server also accepts a username; and the server accepts an unused backup code at sign-in, but the code box takes only 6 characters while backup codes have the form `xxxxx-xxxxx`.
> ⚠️ To be confirmed: password sign-in itself has no attempt limit (only the code step is limited); whether a limit is required.

Change from original BRD: Screen 1 asked for password sign-in only. Two-step verification is now mandatory, sessions are renewed and re-checked continuously, and every failure gives one message.

Expected Result Every registered, active user signs in with password and authenticator code and lands on the dashboard for their role, while wrong details, inactive accounts and guessed codes are refused without revealing which accounts exist.

SCREEN 2: Account Recovery (Forgot your password?)

Field	Value
Screen name	Forgot your password?
Module	Authentication
Access	Everyone (Visitor); usable by Administrator, Manager, Employee and Custom Role users who have two-step verification on
Priority	Must

Purpose Let a user who forgot the password choose a new one without an administrator, proving identity with the current code from the authenticator app.

Screen Elements

- Title **Forgot your password?** with a short explanation.
- Step 1: **Email or username**, **Code from your authenticator app**, **Continue**, and the note "This works if you turned on two-step verification from your Profile. If you did not, ask an administrator to set a new password for you."
- Step 2: "Code verified. Choose a new password for your account.", **New password**, **Confirm new password**, **Reset password**, **Start over** (only when the reset is refused).
- **← Back to sign in** link and legal footer.

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Functional Requirements

ID	Requirement	Priority
REQ-AUTH-19	The system shall let a user prove identity with their email or username and the current 6-digit authenticator code.	Must
REQ-AUTH-20	The system shall offer recovery only for active accounts that have two-step verification on.	Must
REQ-AUTH-21	The system shall not accept a backup code as a recovery factor.	Must
REQ-AUTH-22	The system shall show the same message for every verification failure, so it never reveals whether an account exists or has two-step verification on.	Must
REQ-AUTH-23	The system shall limit verification to 5 attempts per account and 20 per network address within any 5 minutes.	Must
REQ-AUTH-24	The system shall not count wrong recovery codes toward the account's sign-in code lockout, and shall still refuse an account that is already locked for codes.	Must
REQ-AUTH-25	The system shall grant a reset permission that is valid for 10 minutes and can be used once, and cancel the user's other open reset permissions when one is used.	Must
REQ-AUTH-26	The system shall require a new password of 6 to 128 characters, entered twice identically.	Must
REQ-AUTH-27	The system shall end every session of the user when the password is reset.	Must
REQ-AUTH-28	The system shall send the account owner a System notification after a successful reset.	Must
REQ-AUTH-29	The system shall return the user to the Login screen with a success notice after the reset.	Should
REQ-AUTH-30	The system shall offer **Start over** when the reset permission has expired or was already used.	Should

Validations & Messages

Situation	Rule / message
Identifier empty	"Email or username is required."
Code missing or not 6 digits	"Enter the 6-digit code from the app."
Wrong account / code / no two-step verification	"The details are incorrect, or the code has expired. Check them and try again."
Too many attempts	"Too many attempts. Try again in a few minutes."
Verification failed (no server message)	"Could not verify the code. Please try again."
Password length wrong	"Use at least 6 and at most 128 characters."
Passwords differ	"Passwords do not match."
Reset permission used or expired	"This reset link is no longer valid. Start again from the login page."
Reset failed (no server message)	"Could not reset your password. Please try again."

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Situation	Rule / message
Notification to the owner	"Your password was reset with your authenticator app. If this was not you, contact an administrator now."

Change from original BRD: New screen. No password, code or link is sent by e-mail or SMS; the authenticator app is the only recovery factor.

Expected Result A user who forgets the password resets it alone in a few minutes; the old password and every old session stop working at once, and the owner is told in case the reset was not theirs.

SCREEN 3: Security Setup (Let's secure your account)

Field	Value
Screen name	Let's secure your account
Module	Authentication
Access	Administrator, Manager, Employee and Custom Role users who are signed in and have not yet set up two-step verification (anyone else is sent to their dashboard)
Priority	Must

Purpose Mandatory first stop for every user without two-step verification: confirm personal details, connect an authenticator app and save backup codes. Until this is done the rest of TaskFlow stays closed.

Screen Elements

- Progress list **Your details**, **Authenticator app**, **Backup codes** (on small screens "Step N of 3").
- Step 1: "Welcome, (first name)", read-only Employee code, Username, Role, Projects, Managers; **Full name**, **Email** (name part with fixed `@taskflow.com`), **Mobile number**; **Sign out** and **Save and continue**.
- Step 2: **Current password** with Show/Hide, **Show my QR code**; QR code, typed key with **Copy key**, **6-digit code**, **Start over**, **Verify and turn on**.
- Step 3: "Two-step verification is on", 10 backup codes, **Copy all**, **Download**, check box "I have saved my backup codes somewhere safe.", **Open TaskFlow**.
- Loading state and error box "We could not load your details." with **Try again**.

Functional Requirements

ID	Requirement	Priority
REQ-AUTH-31	The system shall require two-step verification for every user by default.	Must
REQ-AUTH-32	The system shall refuse every screen and function except sign-in, the user's own profile and the setup steps while setup is pending, with "Set up two-step verification to continue."	Must
REQ-AUTH-33	The system shall let the user confirm and save full name, email and mobile number before securing the account.	Must
REQ-AUTH-34	The system shall require the current password before showing a new setup key and QR code, limited to 5 attempts in 5 minutes.	Must

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ID	Requirement	Priority
REQ-AUTH-35	The system shall let the user copy the setup key when the QR code cannot be scanned.	Should
REQ-AUTH-36	The system shall turn on two-step verification only after a correct 6-digit code from the app; a backup code is not accepted for this.	Must
REQ-AUTH-37	The system shall refuse to start setup again while two-step verification is on.	Must
REQ-AUTH-38	The system shall create exactly 10 single-use backup codes and show them only once, with copy and download options.	Must
REQ-AUTH-39	The system shall keep **Open TaskFlow** disabled until the user confirms that the backup codes are saved, then open the user's dashboard.	Must
REQ-AUTH-40	The system shall store the setup key encrypted and the backup codes only as hashes.	Must
REQ-AUTH-41	The system shall let the user sign out without finishing setup.	Should

Validations & Messages

Situation	Rule / message
Full name empty (max 200 characters)	"Enter your full name."
Email name part invalid	"Use letters, digits, dot, underscore or hyphen before @taskflow.com."
Email in use	"Email already exists."
Same email and mobile exist	"A user with this email and mobile number already exists."
Mobile number empty	"Enter your mobile number."
Mobile number format	"Use international format, for example +919876543210."
Password empty / wrong	"Enter your password." / "Current password is incorrect."
Too many password attempts	"Too many attempts. Try again in a few minutes."
Code missing or not 6 digits	"Enter the 6-digit code from the app."
Wrong code	"The code is incorrect or has expired."
Code lockout	"Too many wrong codes. Try again in a few minutes."
Setup already on	"Authenticator MFA is already on. Turn it off first to set it up again."
No key yet	"Start the setup first to get a key."
Copy failed	"Could not copy. Select the text and copy it by hand."
Any other page while setup pending	"Set up two-step verification to continue."

> ⚠️ To be confirmed: regenerating backup codes is available in the system but no screen offers it.
> ⚠️ To be confirmed: after an administrator resets an employee's two-step verification (Employee Management), the confirmation text says the employee "can sign in with just their password", while with two-step verification required the employee is sent to Security Setup on the next sign-in.

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Change from original BRD: New screen (two-step verification was not in the original BRD).

Expected Result No one uses TaskFlow until the account is protected by an authenticator app and backup codes.

SCREEN 4: Legal Document Pages

Field	Value
Screen name	Copyright Notice, Terms of Use, Privacy Policy, Disclaimer, Third-Party Licence Notices
Module	Legal
Access	Everyone – Visitor, Administrator, Manager, Employee and Custom Role (no sign-in, no permission)
Priority	Must

Purpose Publish TaskFlow's five legal documents so that everyone knows who owns TaskFlow, the conditions of use, what data is kept and which open-source components it uses.

Screen Elements

- Top bar with "TF" mark, "TaskFlow — Intelligent Task Allocation & Tracking System" and a back link (**Back to the application** or **Back to sign in**).
- Heading: title, one-line summary, "Last updated: 21 October 2026 · © 2026 Jojo George. All Rights Reserved."
- **Contents** box listing the sections (Copyright 8, Terms 14, Privacy 11, Disclaimer 7, Licences 3).
- Document body with paragraphs, lists, notes and tables ("What we collect" in Privacy, "Third-party components" in Licences).
- Ownership line at the foot of the page.

Functional Requirements

ID	Requirement	Priority
REQ-LEGAL-01	The system shall publish five legal documents: Copyright Notice, Terms of Use, Privacy Policy, Disclaimer and Third-Party Licence Notices, each on its own page.	Must
REQ-LEGAL-02	The system shall open the legal pages without sign-in and without any permission.	Must
REQ-LEGAL-03	The system shall show on each page the title, summary, last-updated date, ownership line and a contents list that jumps to each section.	Must
REQ-LEGAL-04	The system shall insert the owner name, year, contact email, rights statement and application name from one owner record into every document.	Must
REQ-LEGAL-05	The system shall send a signed-in user back to their dashboard (Administrator → Administrator Dashboard, Manager → Manager Dashboard, others → personal dashboard) and a visitor back to the Login screen.	Should
REQ-LEGAL-06	The system shall show "Legal document not found" when no document matches the page.	Should
REQ-LEGAL-07	The system shall keep the documents read-only; they cannot be edited from the application and opening them creates no record, notification or mail.	Must
REQ-LEGAL-08	The Privacy Policy shall state what data TaskFlow keeps, why, where it is stored, who can see it, how it is protected and how long it is kept.	Must

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ID	Requirement	Priority
REQ-LEGAL-09	The Terms of Use and Privacy Policy shall state that, when a cloud AI model is chosen, the question and the task details behind it are sent to that provider.	Must
REQ-LEGAL-10	The Disclaimer shall state that assistant output and allocation suggestions are advisory only and must be checked before acting on them.	Must

Validations & Messages

Situation	Rule / message
Document not found	"Legal document not found" / "The document you requested is not available." / "Choose one of the documents from the footer, or back to sign in." (or "back to the application" when signed in)
Back link	"Back to the application" (signed in) / "Back to sign in" (not signed in)

> ⚠️ To be confirmed: an address under `/legal/` that is not one of the five documents is currently sent to the Login screen by the application's catch-all route, so the "not found" message may not be reached.

Change from original BRD: New screens (not in the original BRD).

Expected Result Anyone, signed in or not, can read the five documents, jump between sections and return to the right place.

SCREEN 5: Legal Footer and Ownership Notice

Field	Value
Screen name	Legal footer (part of other screens)
Module	Legal
Access	Everyone – Visitor (Login, Account Recovery) and all signed-in roles (every in-app screen)
Priority	Must

Purpose Show the ownership statement and one-click links to the legal documents at the bottom of every main screen.

Screen Elements

- Ownership line "© 2026 Jojo George. All Rights Reserved. · TaskFlow — Intelligent Task Allocation & Tracking System."
- Links **Copyright**, **Terms of Use**, **Privacy Policy**, **Disclaimer**, **Licences**.
- Browser tab title and page details (author, copyright, description).

Functional Requirements

ID	Requirement	Priority
REQ-LEGAL-11	The system shall show the legal footer with the ownership line at the bottom of the Login screen, the Account Recovery screen and every signed-in screen.	Must

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ID	Requirement	Priority
REQ-LEGAL-12	The system shall show the five document links in the footer, always in the order Copyright, Terms of Use, Privacy Policy, Disclaimer, Licences.	Must
REQ-LEGAL-13	The system shall use the same owner record for the footer and the documents, so the ownership statement is identical everywhere.	Must
REQ-LEGAL-14	The system shall not show the footer on the legal pages, which carry their own ownership line.	Could
REQ-LEGAL-15	The system shall set the browser page title "TaskFlow — Intelligent Task Allocation & Tracking System" and the author, copyright and description details when the application starts.	Should

Validations & Messages

Situation	Rule / message
Ownership line	"© 2026 Jojo George. All Rights Reserved. · TaskFlow — Intelligent Task Allocation & Tracking System."
Validation / no permission	Not applicable (read-only, public).

Change from original BRD: New (not in the original BRD).

Expected Result The ownership statement and the five links appear the same way on every main screen and in the browser page details.

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6. Business Rules

Global rules that apply to every module. Sub-module BRDs add their own rules (`BR-<CODE>-nn`).

ID	Rule	Applies to
BR-01	Only signed-in, active users can use TaskFlow; only Login, Account Recovery and the Legal pages are public.	All screens
BR-02	Deactivated and deleted users cannot sign in, and their open sessions stop working on the next request.	Authentication, Employee Management
BR-03	Two-step verification with an authenticator app is mandatory for every user; until it is set up, every screen except Security Setup and the own profile is closed.	All signed-in screens
BR-04	Access is decided by permissions: every role is a set of permissions from one fixed catalogue, checked by the server on every request; a role change applies on the user's next action.	All modules
BR-05	The system roles Administrator, Manager and Employee cannot be changed; custom roles can. "Always On" permissions (own dashboard, own tasks, own profile) cannot be removed.	Role Configuration
BR-06	Data is shown in the scope the user's permissions allow: Own, Team or Organisation.	Dashboards, tasks, board, reports, workload

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ID	Rule	Applies to
BR-07	Every task has a unique code generated automatically (`TSK-` followed by six digits).	Tasks
BR-08	Every task has an assigned active employee, a project, a start date, a deadline later than the start date and estimated hours greater than 0; deadline and estimated hours are stored separately.	Tasks
BR-09	The assigned employee is always a notification recipient of the task; authorised users may add up to 50 more recipients.	Tasks, reminders
BR-10	Only authorised users can create, edit, reassign, delete or manage recipients of tasks; employees update only tasks they can access.	Tasks, board
BR-11	A task starts as To Do and changes status only along the allowed moves (section 4.1); Completed and Cancelled are final.	Tasks, board
BR-12	A task is overdue when its deadline has passed and it is neither Completed nor Cancelled; overdue is worked out when needed, never stored.	Tasks, dashboards, reports, reminders
BR-13	Completed and cancelled tasks receive no further deadline reminders.	Reminders
BR-14	The reminder schedule is configurable without changing code; the defaults are 2 days and 1 day before the deadline, the deadline day, and daily while overdue.	Reminder Settings, reminders
BR-15	Each reminder (task, person, reminder type and day) is delivered at most once, however often the engine runs or the application restarts.	Reminders
BR-16	Reminders go to the task's recipients and its creator, active users only.	Reminders
BR-17	One application timezone (default Asia/Kolkata) decides which calendar day a deadline falls on for every user.	Tasks, dashboards, reminders, reports
BR-18	A reassignment needs a reason and keeps the task's comments and history.	Tasks
BR-19	Important task changes are recorded in an activity history that ordinary users cannot change or delete.	Tasks
BR-20	No user, including an Administrator, can see another user's notifications, mail or assistant conversation; such records are reported as not found.	Communication, TFA
BR-21	Records that carry history (tasks, employees, projects, departments) are removed by soft delete so that historical information is kept.	Tasks, Management
BR-22	Dashboards, the Task List and reports use the same rules for pending, in progress, completed and overdue.	Dashboards, tasks, reports
BR-23	Messages stay inside TaskFlow: in-app notifications and internal Mail only; nothing is sent by external e-mail or SMS.	Communication, Authentication
BR-24	Every employee email uses the fixed `@taskflow.com` domain and is unique.	Employee Management, My Profile, Security Setup
BR-25	A password change, a password reset and an administrator MFA reset end every session of the user.	Authentication, My Profile, Employee Management
BR-26	The last active Administrator cannot be deactivated, deleted or moved to another role, and nobody can delete their own account.	Employee Management

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ID	Rule	Applies to
BR-27	TFA never acts on its own: every prepared action runs only after the user confirms it, only within the user's own permissions, and every outcome is recorded.	TFA Assistant
BR-28	Passwords, authenticator keys and AI provider keys are never shown again or logged; passwords are hashed and keys encrypted; TFA refuses passwords and keys typed into the chat.	All modules
BR-29	The legal documents and the ownership footer are public and identical for everyone.	Legal

Change from original BRD (rules):

Original rule	Current rule	Change
1	BR-07	Kept; code format `TSK-` plus six digits
2, 3, 4	BR-08	Kept; project and estimate greater than 0 added
5, 6	BR-09, BR-16	Extended: the creator also receives reminders; at most 50 extra recipients
7, 8	BR-10	Kept
9	BR-12	Kept; overdue is derived, never stored
10	BR-13	Kept
11	BR-14	Kept
12	BR-18	Kept; a reason is required
13	BR-19	Kept
14	BR-20	Extended to mail and assistant conversations
15	BR-15	Kept
16	BR-23	Replaced: no external e-mail; in-app notifications are always created and a Mail copy is optional
17	BR-17	Kept
18	BR-02	Kept
19	BR-21	Kept (soft delete)
20	BR-22	Kept
–	BR-01, BR-03 – BR-06, BR-11, BR-24 – BR-29	New

7. Non-Functional Requirements

ID	Category	Requirement	Measure / target
NFR-01	Performance	Long lists are shown in pages.	Tasks and notifications 20 rows per page by default, at most 100; Mail 25 per page.

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ID	Category	Requirement	Measure / target
NFR-02	Performance	Dashboards and reports show totals calculated by the server, not by the browser.	⚠️ To be confirmed: a response-time target (for example seconds per screen) has not been set.
NFR-03	Performance	The CSV export of a report is limited.	At most 50,000 rows per export.
NFR-04	Security	Passwords are stored only as salted hashes; new passwords have 6 to 128 characters.	No plain-text password stored, shown or logged.
NFR-05	Security	Two-step verification (time-based 6-digit codes, 10 single-use backup codes) is required for everyone and enforced by the server, not only by the screens.	No function except sign-in, own profile and setup works before setup.
NFR-06	Security	Sessions are short and renewable.	Access pass 30 minutes; renewal cookie 7 days, HttpOnly, Secure, SameSite Strict, replaced at each renewal.
NFR-07	Security	Guessing is throttled.	5 wrong codes lock for 5 minutes; recovery 5 per account and 20 per address in 5 minutes; current-password re-entry 5 per 5 minutes; TFA 20 messages per minute per user.
NFR-08	Security	All input is validated by the server and database queries are safe against injection.	Every write is validated; invalid input returns field messages.
NFR-09	Security	Protection against unauthorised record access.	Another user's task, mail, notification or conversation is refused or reported as not found.
NFR-10	Security	Secrets are protected.	AI provider keys and authenticator keys encrypted with a key ring kept outside the database; only the last 4 characters of a key are ever shown.
NFR-11	Privacy	Data stays on the organisation's server unless a cloud AI model is chosen.	Users see a notice the first time a cloud model is used; the local model is the default.
NFR-12	Usability	Clear, consistent screens for users with basic computer knowledge.	Every refused action shows a message; destructive actions and every TFA action ask for confirmation; menus show only what the user may use.
NFR-13	Reliability	Tasks, history, notifications and reminder history are stored permanently.	Data survives a restart of the application.
NFR-14	Reliability	The reminder engine resumes after a restart without duplicates.	First run at start-up, then every 60 seconds; a failed run is logged and the next run continues; no reminder delivered twice.
NFR-15	Maintainability	The application is modular and layered with separate authentication, tasks, notifications, reports and data logic, and the structure is checked automatically.	Structure rules run with the automated tests.

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ID	Category	Requirement	Measure / target
NFR-16	Compatibility	Works in modern desktop browsers with a responsive layout for smaller screens.	Voice input needs Chrome or Edge; typing always works. The server runs on Windows, Linux and macOS.
NFR-17	Backup & Recovery	All stored data lives in one data folder with a documented backup and restore procedure.	Database and mail attachments backed up together; encryption key ring backed up separately.
NFR-18	Accessibility	Screens can be used with the keyboard and are readable.	⚠️ To be confirmed: no accessibility target has been defined.
NFR-19	Observability	Every request is logged without its content.	Method, path, status, time, user and trace id; refused authorisations and TFA action outcomes are recorded.
NFR-20	Offline operation	TaskFlow works without the internet.	Notifications, Mail and the local AI model need no external service; only cloud AI providers need the internet.

8. Data Requirements

Entity / data set	Purpose	Key fields	Retention / rules
Users	Sign-in accounts and employee details	Employee code, full name, username, email, mobile number, role, active flag, last sign-in	Soft delete; email on `@taskflow.com` and unique; username unique
Security records	Sessions, password reset permissions, authenticator key, backup codes	Renewal token, expiry, used/revoked time; encrypted key; hashed codes	Tokens and codes stored only as hashes; reset permission 10 minutes, single use
Roles and permissions	Access control	Role name, system flag, granted permissions	Permission catalogue fixed; system roles read-only; a role in use cannot be deleted
Departments	Group people; source of managers	Name, description, managers, employees	Soft delete; unique name; cannot be deleted while it has members
Projects, positions and allocations	Teams and time share	Project name, description, members, position, allocation %, project-manager flag	Soft delete; at most 100% per person across projects; every team has a project manager
Tasks	Work items	Code, title, description, project, assignee, creator, priority, start date, deadline, estimated and worked hours, status, reminder days, overdue frequency, completed and cancelled time	Soft delete; code unique; overdue derived
Task recipients	Who is notified about a task	Task, person	Assignee always included; at most 50 more

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Entity / data set	Purpose	Key fields	Retention / rules
Task comments	Discussion	Task, author, text, created and edited time	Soft delete; only the author edits or deletes
Task activity history	Record of changes	Task, event type, user, old value, new value, message, time	Append-only; not editable by ordinary users
Notifications	In-app alerts	Person, task, type, message, read flag, time	Never deleted, only marked read; private to the owner
Reminder history	Duplicate prevention	Task, recipient, reminder type, scheduled day, status, processed time	One row per task, person, type and day
Internal mail	Messages between users	Conversation, sender or "TaskFlow", subject, body, recipients (To/Cc), per-person folder and flags, attachments	Private per mailbox; attachments stored in the data folder
Application settings	Organisation-wide configuration	Name, timezone, default priority, notifications, reminder defaults, editor, time	Non-secret values only; editor and time recorded
AI provider settings and models	TFA configuration	Provider, enabled flag, model, server address, encrypted key, last 4 characters	Keys encrypted, never returned
Assistant conversations, actions and audit	TFA history and safeguards	Person, messages, prepared action, status, expiry, outcome	Own conversation only; actions expire after 10 minutes; one audit record per outcome

9. Integrations & External Dependencies

System / service	Purpose	Required?	Notes
Authenticator app (for example Google Authenticator, Microsoft Authenticator, Authy)	Two-step verification codes and account recovery	Required for every user	Works offline on the user's phone; lost phone → backup codes or administrator MFA reset
Ollama (local AI server)	TFA answers on the organisation's own server	Optional	On by default for new installations; nothing leaves the server. Without it TFA shows a provider message and the rest of TaskFlow works
Google Gemini, OpenAI GPT (cloud AI)	Alternative TFA models	Optional	Need an API key entered in TFA Settings and the internet; questions and task details are sent to the provider
Browser speech recognition and speech output	TFA voice input and read-aloud	Optional	Chrome or Edge; no audio is stored by TaskFlow
Local file storage (data folder)	Database, mail attachments, encryption key ring	Required	Created at start-up; the application stops with a clear message if it cannot write there

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System / service	Purpose	Required?	Notes
External e-mail (SMTP)	–	Not used	Original Screen 14 superseded by internal Mail

10. Reporting Requirements

Report	Audience	Filters	Output formats
Task Reports – total, completed, pending, overdue and completion summary	Administrator, Manager, custom roles with `reports.view`	Deadline range, employee, project, status, priority	Screen; CSV (`reports.export`)
Dashboard analytics – by status, priority, deadline, time, department, employee, manager, project	Administrator (organisation), Manager (team), every user (own)	Dashboard filters and period chips	Screen, with drill-down to the Task List
Employee Workload – active, high-priority, overdue, completed per employee, projects and allocation	Administrator, Manager	Project, employee	Screen
Activity History – task changes	Users with `tasks.activity.view` who can access the task	Activity filters	Screen

Details: BRD-Reports, BRD-Dashboard, BRD-Employee-Workload, BRD-Task-Management.

11. Recommended Technology Stack (optional)

> A BRD states *what* is needed; this section is a recommendation. The final choice is recorded in the TDD.

Component	Recommended technology	Reason
Backend	.NET 10, ASP.NET Core Web API, FluentValidation	Typed, layered, server-side validation and permission checks
Frontend	Angular 19 (TypeScript 5.7, RxJS), Chart.js with ng2-charts, Angular CDK (drag and drop), qrcode	Single-page application with charts, a drag-and-drop board and QR codes for setup
Database	SQLite through Entity Framework Core 10	Single file, easy to back up, schema created automatically
Authentication	Token-based sessions with a renewal cookie; password hashing; time-based authenticator codes	Secure sessions and mandatory two-step verification
Scheduled reminders	Background service inside the application	No external scheduler; resumes after restart
AI assistant	Ollama (local) by default; Google Gemini or OpenAI GPT optional	Free local option; cloud by choice

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Component	Recommended technology	Reason
Testing	xUnit unit and integration tests; Jasmine and Karma for the frontend	Automated checks of rules, endpoints and structure

Change from original BRD: The original recommended Python Flask, Bootstrap, SQLAlchemy, APScheduler, SMTP and Pytest. SQLite and Chart.js are kept; the rest was replaced by the stack above, and SMTP was dropped.

12. Assumptions, Constraints & Dependencies

Type	Description
Assumption	TaskFlow serves one organisation; users have a modern desktop browser and a phone with an authenticator app.
Assumption	All deadlines are compared in one application timezone (default Asia/Kolkata).
Assumption	Reminder defaults are first reminder 2 days, second 1 day, deadline-day reminder on, overdue daily; each task can choose its own values when it is created.
Assumption	Development installations may create three demonstration accounts (Administrator, Manager, Employee); this is off by default elsewhere.
Constraint	Single application server and a single SQLite database file; sign-in throttles and the assistant rate limit are kept in memory and reset on restart.
Constraint	All stored data lives in one data folder (default `C:\TaskFlow` on Windows).
Constraint	Every email address uses the `@taskflow.com` domain.
Dependency	Authenticator app on each user's phone.
Dependency	Optional: Ollama for the local assistant; internet and an API key for cloud AI providers.

13. Risks

ID	Risk	Impact	Likelihood	Mitigation
R-01	Reminders run twice if more than one server instance is started.	Medium	Low	Database prevents duplicate reminders; single instance assumed.
R-02	Loss of the encryption key ring.	High	Low	Back up the key ring separately; otherwise AI keys must be re-entered and every user needs an MFA reset.
R-03	A user loses phone and backup codes.	Medium	Medium	Administrator MFA reset ends sessions and lets the user set up again.
R-04	Task details leave the server when a cloud AI model is used.	Medium	Medium	Local model is the default; notice on first cloud use; passwords and keys are blocked in chat.
R-05	Wrong "due today" or "overdue" because of different timezones.	Medium	Low	One application timezone on the server; unknown timezone falls back to UTC so reminders never stop.

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ID	Risk	Impact	Likelihood	Mitigation
R-06	Password guessing at sign-in, which has no attempt limit.	Medium	Low	Code step required for every account; ⚠️ To be confirmed whether a password attempt limit is needed.
R-07	Upgrades that change data irreversibly (for example email domain change).	High	Low	Back up before every upgrade.
R-08	Slow writes when many users work at the same time on one SQLite file.	Medium	Low	Short transactions; reminder runs skip already delivered reminders before writing.
R-09	Managers see organisation-wide reports and workload, wider than their team.	Medium	Medium	Documented in BRD-Reports and BRD-Employee-Workload; ⚠️ To be confirmed whether a team limit is required.

14. Acceptance Criteria

ID	Criterion	Verified by
AC-01	Administrator, Manager, Employee and custom-role users sign in and land on the dashboard for their role.	SCREEN 1, REQ-AUTH-01, REQ-AUTH-08 – REQ-AUTH-12; AC-DASH-01
AC-02	A wrong password, an unknown account and an inactive or deleted account all get "Invalid email or password." and no session.	SCREEN 1, REQ-AUTH-02, REQ-AUTH-03; AC-EMP-03
AC-03	With two-step verification on, a session opens only after a valid, unused code; a reused code is refused; 5 wrong codes lock code entry for 5 minutes; a sign-in step older than 5 minutes is refused.	SCREEN 1, REQ-AUTH-04 – REQ-AUTH-07
AC-04	A session survives a reload, renews itself, applies role changes within 15 seconds, ends on Sign out and after a password or MFA reset, and the user then sees the matching notice.	SCREEN 1, REQ-AUTH-13 – REQ-AUTH-18; AC-ROLE-11
AC-05	A user with two-step verification resets a forgotten password with an authenticator code; every failure shows the same message; the 6th attempt in 5 minutes is refused; the reset permission expires after 10 minutes; old sessions end and the owner is notified.	SCREEN 2, REQ-AUTH-19 – REQ-AUTH-30; AC-NOTIF-09
AC-06	A new user cannot open any screen until Security Setup is finished; setup needs the current password and a valid code; 10 backup codes are shown once; Open TaskFlow is enabled only after confirming they are saved.	SCREEN 3, REQ-AUTH-31 – REQ-AUTH-41; AC-PROF-12, AC-PROF-13
AC-07	Administrators manage employee accounts, including activation, deletion protections and MFA reset.	AC-EMP-01 – AC-EMP-13
AC-08	Managers create and assign tasks with a unique code; the assignee is notified.	AC-TASK-06, AC-TASK-09
AC-09	Employees view their tasks and update them: status (allowed moves only), worked hours and comments.	AC-TASK-01, AC-TASK-12, AC-TASK-13, AC-TASK-16, AC-BOARD-03
AC-10	Upcoming, due-today and overdue tasks are shown correctly in the application timezone, with the same figures on dashboards, list and reports.	AC-TASK-03, AC-DASH-05, AC-DASH-12, AC-RPT-02

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ID	Criterion	Verified by
AC-11	Deadline reminders are generated automatically on the default schedule.	AC-NOTIF-11, AC-REM-08
AC-12	The assignee, every recipient and the creator receive each reminder; inactive users receive none.	AC-NOTIF-12
AC-13	In-app reminder notifications are always created when reminders run; a Mail copy is added only when "Also deliver reminders to Mail" is on.	AC-NOTIF-15, AC-REM-05
AC-14	Completed and cancelled tasks stop receiving reminders.	AC-NOTIF-14, AC-TASK-12
AC-15	Duplicate reminders are prevented, also after a restart.	AC-NOTIF-13
AC-16	Task reassignment requires a reason and preserves comments and history.	AC-TASK-15
AC-17	Managers view workload summaries with high-load highlighting.	AC-WORK-01 – AC-WORK-04
AC-18	Reports can be filtered and exported to CSV.	AC-RPT-05, AC-RPT-11
AC-19	Unauthorised access is prevented: menus are hidden, addresses redirect to the dashboard, the server refuses the request, and other users' notifications and mail are reported as not found.	AC-EMP-02, AC-RPT-01, AC-REM-11, AC-NOTIF-07, AC-MAIL-06
AC-20	After a restart of the server, tasks, history and reminder history are preserved and reminders continue without duplicates.	AC-NOTIF-13, AC-NOTIF-16
AC-21	Reminder defaults, timezone, default priority, notifications and roles are changed from Settings without code changes and take effect without restart.	AC-REM-09, AC-APP-05, AC-ROLE-05
AC-22	Users exchange internal mail privately; nothing is sent outside TaskFlow.	AC-MAIL-06, AC-MAIL-13
AC-23	TFA prepared actions run only after confirmation and within the user's permissions, and every outcome is audited.	AC-TFA-09, AC-TFA-10, AC-TFA-17
AC-24	Anyone, signed in or not, opens each of the five legal documents from the footer, uses the contents list and returns with the back link; opening them creates no record.	SCREEN 4, REQ-LEGAL-01 – REQ-LEGAL-07
AC-25	The Privacy Policy, Terms of Use and Disclaimer contain the data, cloud-AI and advisory statements.	SCREEN 4, REQ-LEGAL-08 – REQ-LEGAL-10
AC-26	The footer with the same ownership line and five links in fixed order appears on Login, Account Recovery and every signed-in screen, not on the legal pages, and the browser tab shows the TaskFlow title.	SCREEN 5, REQ-LEGAL-11 – REQ-LEGAL-15

Change from original BRD: Original acceptance criteria 1–15 map to AC-01 and AC-07 – AC-20. Original criterion 8 ("in-app notifications remain available when email delivery fails") is now AC-13. AC-02 – AC-06 and AC-21 – AC-26 are new.

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15. Requirements Traceability Matrix

Objective	Business rule(s)	Screen(s)	Functional requirement(s)	Acceptance criterion
OBJ-01	BR-07, BR-08, BR-09, BR-10	Screen map 8–10	REQ-TASK-* (BRD-Task-Management)	AC-08
OBJ-02	BR-11, BR-12, BR-22	Screen map 4–9, 13	REQ-TASK-, *REQ-BOARD-*, REQ-DASH-*	AC-09, AC-10
OBJ-03	BR-12, BR-13, BR-14, BR-15, BR-17	Screen map 18, 21	REQ-NOTIF-, *REQ-REM-*	AC-11, AC-14, AC-15, AC-20
OBJ-04	BR-09, BR-16, BR-23	Screen map 8, 18, 19	REQ-TASK-, *REQ-NOTIF-*, REQ-MAIL-*	AC-12, AC-13
OBJ-05	BR-06, BR-12, BR-22	Screen map 4–6, 14	REQ-DASH-, *REQ-WORK-*	AC-10, AC-17
OBJ-06	BR-20, BR-23	Screen map 11, 18, 19	REQ-TASK-, *REQ-NOTIF-*, REQ-MAIL-*	AC-22
OBJ-07	BR-18, BR-19, BR-21	Screen map 10, 12	REQ-TASK-*	AC-16
OBJ-08	BR-06, BR-22	Screen map 20	REQ-RPT-*	AC-18
OBJ-09	BR-01, BR-02, BR-03, BR-25, BR-28	SCREEN 1	REQ-AUTH-01 – REQ-AUTH-18	AC-01, AC-02, AC-03, AC-04
OBJ-09	BR-03, BR-23, BR-25	SCREEN 2	REQ-AUTH-19 – REQ-AUTH-30	AC-05
OBJ-09	BR-03, BR-24, BR-28	SCREEN 3	REQ-AUTH-31 – REQ-AUTH-41	AC-06
OBJ-09	BR-04, BR-05, BR-06, BR-20, BR-24, BR-26	Screen map 15–17, 24, 25	REQ-EMP-, *REQ-DEPT-*, REQ-PROJ-, *REQ-ROLE-*, REQ-PROF-*	AC-07, AC-19
OBJ-10	BR-04, BR-05, BR-14, BR-17	Screen map 21–24	REQ-REM-, *REQ-APP-*, REQ-TFAS-, *REQ-ROLE-*	AC-21
OBJ-11	BR-27, BR-28	Screen map 23, 26	REQ-TFA-, *REQ-TFAS-*	AC-23
OBJ-12	BR-01, BR-29	SCREEN 4	REQ-LEGAL-01 – REQ-LEGAL-10	AC-24, AC-25
OBJ-12	BR-29	SCREEN 5	REQ-LEGAL-11 – REQ-LEGAL-15	AC-26

`REQ-<CODE>-*` stands for the requirements of the sub-module BRD with that code (section 3.4); their detailed traceability is in section 15 of each sub-module BRD.

16. Project Deliverables

#	Deliverable	Format
1	Complete application source code (backend API and Angular frontend)	Repository
2	Database schema (created by migrations at start-up) and initial data (permission catalogue, system roles, AI models)	Code-first migrations and seeders

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#	Deliverable	Format
3	BRD (overall and 16 sub-modules), FRD, Development Plan, module workflow docs, TDD, HLD, LLD	Markdown, DOCX, PDF
4	Installation and setup instructions, administrator account setup, backup and restore procedure	README (Markdown)
5	Test cases and test results	Automated unit and integration tests and their reports
6	Sample data for demonstration (three development accounts)	Seed option in development configuration
7	Legal documents (Copyright, Terms, Privacy, Disclaimer, Licences)	Pages inside the application
8	Demonstration-ready application	Running installation

17. Glossary

Term	Meaning
Assignee	The employee a task is assigned to; always a notification recipient.
Recipient	Any user configured to receive notifications and reminders for a task.
Creator	The user who created a task; also receives its reminders.
Two-step verification (MFA)	Sign-in that needs the password and a 6-digit code from an authenticator app.
Backup code	One of 10 single-use codes, form `XXXXX-XXXXX`, for use when the phone is not at hand.
Security Setup	The mandatory first-time screen that turns on two-step verification.
Reset permission	Single-use permission, valid 10 minutes, to set a new password after account recovery.
Session	The signed-in state, made of a 30-minute access pass and a 7-day renewal cookie.
Security stamp	Value that changes after a password change, reset or MFA reset and ends all old sessions.
Permission	A named right from the fixed catalogue (for example `tasks.create`).
Always On	A permission every role holds that cannot be removed.
Custom role	A role created by an administrator from the permission catalogue.
Scope (Own / Team / Organisation)	How much data a user sees: own records, the team's records, or everything.
Team	For a Manager: active people in a department they manage or on a project where they are project manager.
In QA	Task status for work that is finished and being checked.
Overdue	Deadline passed and task neither Completed nor Cancelled; never stored.
Reminder engine	Background process that sends deadline reminders every 60 seconds.
Internal Mail	Private mailbox between TaskFlow users; replaces external e-mail.
TFA	TaskFlow Assistant, the AI assistant panel.

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Term	Meaning
Action Card	TFA card showing a prepared action that runs only after **Confirm**.
Application timezone	The single timezone used for all deadline days (default Asia/Kolkata).

18. Project Summary

TaskFlow – Intelligent Task Allocation & Tracking System brings task assignment, progress tracking, automatic deadline reminders, workload monitoring, internal communication and reporting into one web application. Its main business value is that every task has one owner, one deadline and one recorded history, and that everyone involved is reminded on time, exactly once, without manual follow-up. Access is protected by mandatory two-step verification and permission-based roles, the organisation configures reminders, timezone, roles and its AI assistant from the screen, and the legal documents make ownership and data handling clear to every user. The key workflows are the task lifecycle, the automatic deadline reminders and secure sign-in with two-step verification; each module's detail is in the 16 sub-module BRDs indexed in section 3.4.

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